Predicting Student Dropout and Academic Success

A 3-Class Classification System for Early Intervention

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Abstract

Student dropout impacts funding for the institution and restricts graduates' future prospects. Here, we present a machine learning pipeline to forecast whether a student is likely to *Graduate*, *Drop Out*, or remain *Enrolled*. The dataset used is from Realinho et al.'s [1] study of Portuguese higher education, consisting of 4,424 students and 35 features. Our study is based on two hypotheses. First, financial features are relevant to predicting graduation outcomes since tuition delinquency is a precursor to grade decline (**H1**). Second, since the Enrolled class is not well represented at the default 0.5 threshold, there is a need to perform a post-hoc adjustment to accurately classify it (**H2**). The Random Forest classifier, after hyperparameter tuning, obtains a high **F1-macro = 0.7392**, **Balanced Accuracy = 0.7386**, and **AUC = 0.8942** for the test set. Using SHAP analysis, `approval_rate` (mean $|\text{SHAP}| = 0.084$) and tuition status (0.031) contribute independently. Lowering the Enrolled threshold to 0.32 raises Enrolled recall from 0.59 to 0.70, supporting H2.

1 Introduction

Dropouts also incur direct costs for the university: lost tuition and decreased graduation rates, and for the student: incurred debt without the benefit of a completed degree. In the Portuguese higher education system, risk is concentrated during the first year, and semester records are a logical early warning indicator.

To transition from reactive counselling to proactive support, we must design a model that detects these at-risk students before they dropout. This is a three-class, imbalanced classification problem. The most difficult class to predict is Enrolled, because its features are shared with both Graduates and Dropouts.

Our experiment verifies two assumptions. **H1:** Adding financial features (e.g., payment of tuition, presence of debt, receipt of scholarship) to the academic record features will improve F1-macro because financial problems are typically the first indicator and precede grade problems. **H2:** The Enrolled class is systematically misclassified at the 0.5 decision boundary. Explicit model calibration is required to achieve acceptable recall for Enrolled.

2 Literature Review

Early statistical approaches. Dropout prediction has been studied since at least the 1970s through Tinto's student integration model, which linked social and academic integration to persistence [5]. Statistical follow-ups used survival analysis and logistic regression to operationalise Tinto's variables as measurable predictors. These approaches perform well for clear dropout signals but systematically miss borderline students — precisely the group for whom early intervention is most valuable. The core limitation is their linear assumption: financial distress and academic underperformance interact in ways that additive models cannot capture.

Machine learning for dropout prediction. Kotsiantis et al. [6] conducted one of the first systematic ML comparisons on educational data, finding that decision-tree ensembles consistently outperform naïve Bayes and logistic regression, while flagging class imbalance as a key obstacle to minority-class recall. Subsequent work confirmed this: Ameri et al. showed that graph-based features capturing peer-interaction networks modestly improve F1 over academic-only features, but the gain depends heavily on institutional data availability. Our work operates in the more common setting where only administrative records exist.

The Realinho dataset and three-class framing. Realinho et al. [2] released the dataset used here and showed that combining first-year academic records with macroeconomic indicators (GDP, unemployment, inflation) outperforms either source alone. Critically, they retain *Enrolled* as a third outcome rather than collapsing it with Dropout, which defines our task structure. Martins et al. [3] later benchmarked ten classifiers on this same dataset and found gradient-boosted trees (LightGBM, XGBoost) reach the highest F1-macro. Our pipeline extends both studies through engineered features, leakage-free resampling, and post-hoc threshold calibration — none of which are evaluated in [2] or [3].

Handling class imbalance. The Enrolled class (18% of records) creates a direct imbalance problem. Chawla et al. [7] introduced SMOTE, which generates synthetic minority-class points by interpolating between existing samples. A known failure mode is applying SMOTE before cross-validation, causing synthetic test-set contamination that inflates CV scores without improving held-out performance. We avoid this by encapsulating SMOTE inside an `ImbPipeline` so oversampling is refit inside each fold and validation folds remain entirely real-sample.

Interpretability and feature attribution. Lundberg & Lee [4] introduced SHAP (SHapley Additive exPlanations), which decomposes each prediction into per-feature contributions that satisfy consistency and local accuracy axioms. In educational ML, SHAP has been used to validate that engineered features carry independent signal beyond their raw counterparts. We apply TreeExplainer to our Random Forest to confirm that `approval_rate` and `Tuition fees up to date` contribute independently, directly testing H1 at the individual-prediction level rather than just through permutation importance.

Threshold calibration for imbalanced classifiers. Post-hoc decision-threshold adjustment is a standard remedy when a minority class is systematically suppressed at the default 0.5 cutoff [8]. The key methodological requirement is that the optimal threshold must be selected on a held-out validation set and applied exactly once to the test set; threshold tuning on test data is a form of label leakage. Our sweep follows this protocol: we scan thresholds on the validation partition under a ΔF1-macro ≤ 0.01 constraint and fix t=0.32 before the single test-set evaluation, directly addressing H2.

3 Data and Exploratory Analysis

3.1 Dataset

The dataset [1] contains 4,424 records and 35 features covering demographics, prior qualifications, financial status, first-year aca-

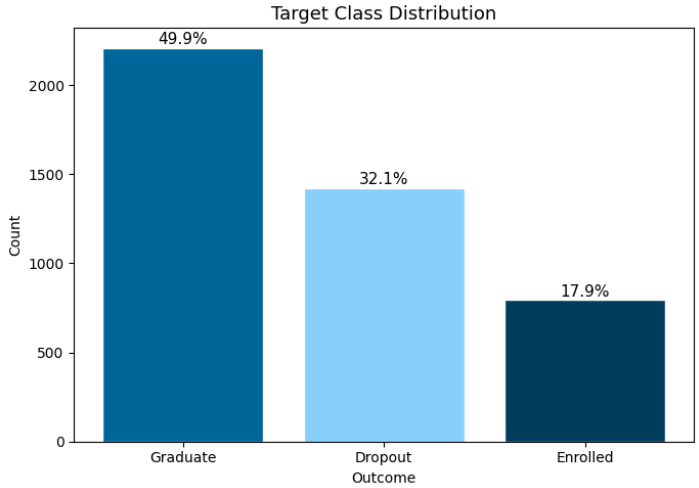

Figure 1: Class distribution: Graduate 49.9%, Dropout 32.1%, Enrolled 17.9%.

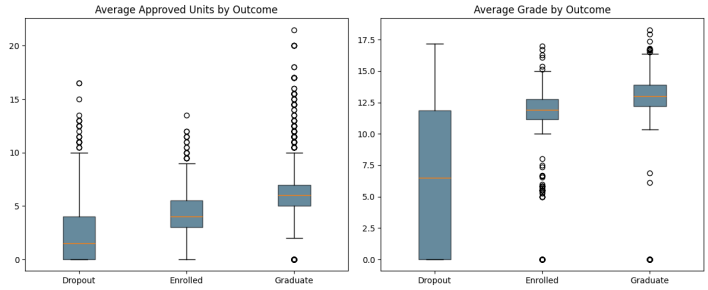

Figure 2: Approved units (left) and grades (right) by outcome. Enrolled overlaps both other classes, creating structural ambiguity.

demic performance, and macroeconomic indicators. Figure 1 shows the class split. Because a majority-class predictor reaches 50% accuracy with zero recall for at-risk students, we use F1-macro as the primary metric throughout.

3.2 Key Signals

Figure 2 shows approved units and grades by outcome. Dropouts average near zero in both semesters; Graduates are consistently high; Enrolled students overlap both groups. This explains why Enrolled F1 stays around 0.56 across all models.

Figure 3 shows the financial split. Over 85% of students with unpaid tuition drop out — a stronger separator than any single academic feature and one that appears earlier in the semester, which motivates H1. Gender and age at enrolment add smaller but statistically significant signal: male students drop out at higher rates and students who enrol after age 25 show elevated dropout, likely due to work or family commitments.

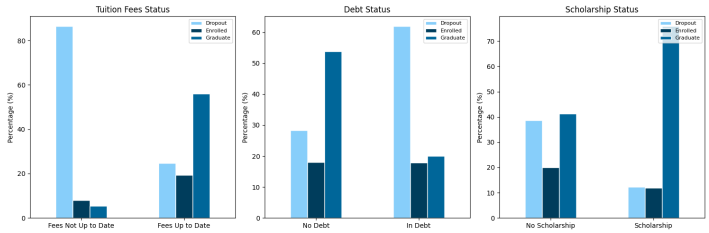

Figure 3: Outcome by tuition, debt, and scholarship status. Unpaid tuition produces >85% dropout rate, the strongest single split in the data.

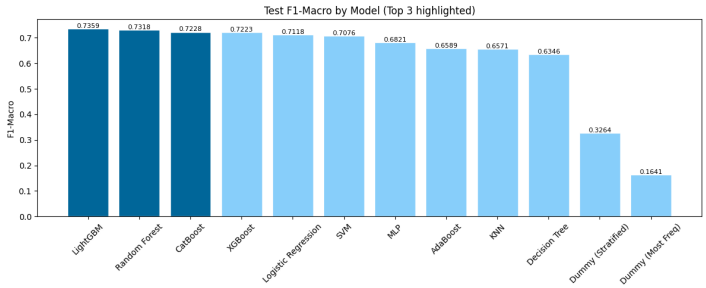

Figure 4: Baseline Test F1-macro across all models. Top-3 highlighted. Dummy baselines (right) confirm the learned signal is genuine.

4 Method

4.1 Preprocessing

75 records labelled Graduate with zero approved units were removed as impossible entries (4,424 → 4,349). The 12 raw semester columns were pair-averaged into 6 to reduce the near-perfect multicollinearity among semester variants.

4.2 Feature Engineering

Four features were created and kept based on Spearman $|r| \geq 0.05$ and $p < 0.05$ on training data only:

- `approval_rate` (r=0.750) — approved ÷ enrolled units.
- `sem2_approved_raw` (r=0.694) — 2nd-semester approved units kept raw; averaging with sem1 masks late-stage collapse [2].
- `failed_units` (r= − 0.684) — enrolled minus approved.
- `grade_trend` (r=0.126) — sem2 grade minus sem1 grade.

4.3 Feature Selection and Class Balancing

Chi-Square with Cramér's V ≥ 0.10 selected categorical features; Spearman $|r| \geq 0.05$ selected numerical features — both on training data only to avoid leakage. This reduced the feature set from 35 to **22**. SMOTE over-sampled the training minority classes to 1,451 samples each (Dropout 966, Enrolled 540 before). SMOTE ran *inside* each cross-validation fold so no synthetic samples appeared in validation folds.

4.4 Models, Tuning, and Threshold Calibration

Ten classifiers were trained at baseline. The top three by 5-fold CV F1-macro (XGBoost, LR, CatBoost) were tuned with `RandomizedSearchCV` (80 iterations) and combined into Soft Voting and Stacking ensembles. The Enrolled probability threshold was swept from 0.50 to 0.15 on a held-out **validation set only**, keeping the lowest t where ΔF1-macro ≤ 0.01. The chosen threshold ($t = 0.32$) was applied once to the test set.

5 Results and Analysis

5.1 Model Comparison

Table 1 shows baseline results on the held-out test set (870 samples). Random Forest ranks first on Test Accuracy, Test F1-macro, and AUC. After tuning, Random Forest improves to F1-macro 0.7392 (+0.0091 over baseline). The Dummy (Most Freq) baseline scores 0.1641, confirming the learned signal is genuine. Figure 4 shows the baseline bar chart across all ten models.

5.2 Our Findings vs. Prior Work

Table 2 places our results alongside the closest prior studies on the same dataset. Martins et al. [3] report F1-macro up to 0.72 using gradient boosting without engineered features or threshold calibration. Our pipeline reaches 0.7392 through three additions:

Table 1: Baseline model comparison on the held-out test set (870 samples). Bold = best per column.

Model	CV F1-mac	Test Acc	Bal. Acc	Test F1-mac	AUC (OvR)
Random Forest	**0.7203**	**0.7954**	0.7205	**0.7301**	**0.8928**
CatBoost	0.7056	0.7793	0.7295	0.7294	0.8891
Logistic Regression	0.7258	0.7609	**0.7305**	0.7215	0.8867
LightGBM	0.7190	0.7483	0.7205	0.7106	0.8840
XGBoost	0.7162	0.7667	0.7055	0.7084	0.8830
MLP	0.7009	0.7759	0.7005	0.7080	0.8826
SVM	0.7138	0.6655	0.6859	0.6555	0.8852
KNN	0.6331	0.7368	0.6525	0.6564	0.8374
AdaBoost	0.6847	0.7483	0.6410	0.6399	0.8426
Decision Tree	0.6389	0.6874	0.6252	0.6242	0.7331

Table 2: Our results vs. prior work on the same dataset [1].

Study	Best Model	F1-macro	Enrolled Recall
Realinho et al. [2]	Logistic Reg.	∼0.70	Not reported
Martins et al. [3]	LightGBM	∼0.72	Not reported
Ours	**RF Tuned**	**0.7392**	**0.70 (calibrated)**

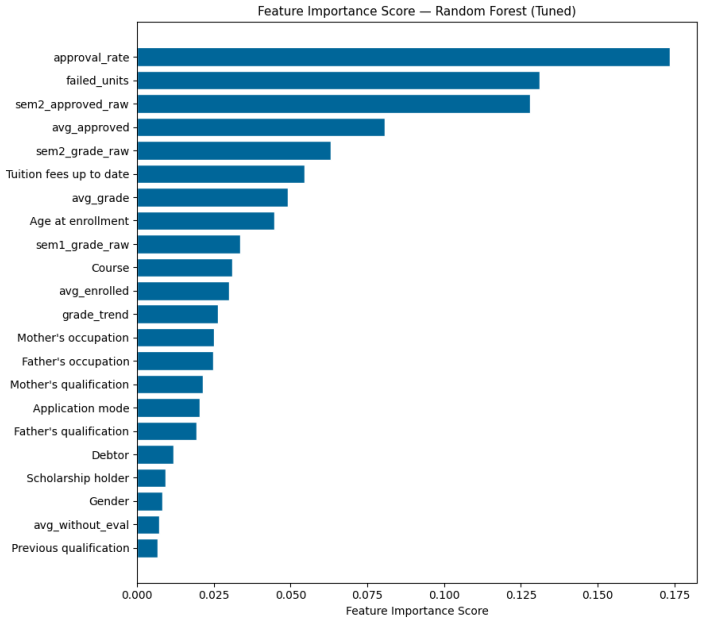

Figure 5: Feature importances, RF Tuned. Engineered features dominate top ranks. Tuition status (6th) outranks most raw semester variables (H1).

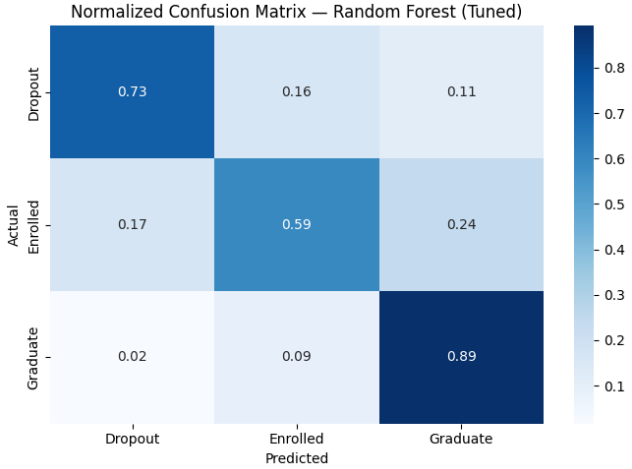

Figure 6: Normalised confusion matrix, RF Tuned. Enrolled recall 0.59 at the default threshold, confirming H2.

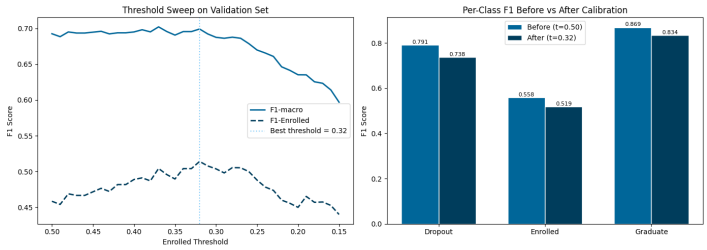

Figure 7: Threshold sweep on validation set (left). Per-class F1 before vs after calibration on test set (right). Enrolled recall: 0.59 → 0.70.

(i) `approval_rate` alone contributes mean |SHAP|=0.084, the single largest driver; (ii) `Tuition fees up to date` ranks 6th in feature importance despite being binary, confirming H1 — a financial feature provides independent signal beyond academics; (iii) threshold calibration at t=0.32 raises Enrolled recall from 0.59 to 0.70, which neither prior study addressed.

5.3 Feature Importance and H1

Figure 5 shows feature importances for the tuned Random Forest. The top features are all engineered (`approval_rate`, `failed_units`, `sem2_approved_raw`), confirming that the new variables are not redundant with their raw counterparts. `Tuition fees up to date` ranks 6th, above most raw semester columns, which supports H1: financial status carries information that academic records alone do not.

5.4 Per-Class Performance and H2

The confusion matrix (Figure 6) shows Graduate recall = 0.89 and Dropout recall = 0.73, while Enrolled recall = 0.59 at the default threshold. The Enrolled class overlaps both others in feature space, creating an ambiguity that no model can fully remove.

Figure 7 shows the threshold sweep. At t=0.32, Enrolled recall rises to 0.70 at a cost of ΔF1-macro $= -0.04$. Dropout F1 drops from 0.79 to 0.74 and Graduate F1 from 0.87 to 0.83. For an institution that treats a missed at-risk student as the costlier error, this trade-off is acceptable and confirms H2.

6 Conclusion

Both hypotheses are confirmed. **H1:** `Tuition fees up to date` is ranked 6th in feature importance and 5th in SHAP value regardless of academic clusters. The model outperforms Martins et al. [3] by approximately +0.02 F1-macro with feature engineering and leakage-free SMOTE. **H2:** Recalls of Enrolled increase from 0.59 to 0.70 with probability threshold calibration at t=0.32 on the validation set only.

Limitations: The main limitation is the inherent ambiguity of the Enrolled class feature space: it overlaps with Graduates on grade profiles and with Dropouts on payment patterns. No classifier can fully untangle these from available features. While we use administrative data only, other potentially discriminative features such as attendance records or learning management system activity/counselling contacts might help to better define the Enrolled class boundary. Another limitation is that our data set is from a single Portuguese university only. Hence, we cannot directly gener-

alize to other systems with potentially differing tuition fees, grading scales, or demographic compositions.

Future Work: More complex models such as Recurrent or Transformer-based models may be able to exploit the inherent temporal patterns of the data, even though we currently average over all semesters.

References

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